



AJKD Atlas of Renal Pathology: Membranous Nephropathy

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Clinical and Pathologic Features

Membranous nephropathy (MN) is an immune complex disease caused by subepithelial deposits. Primary MN is a common cause of nephrotic syndrome. About a third of patients reach remission, a third are stable, and a third have progressive loss of kidney function and persistent proteinuria. The phospholipase A₂ receptor (PLA₂R), expressed on podocytes, is the antigen in about 70% of patients with primary MN. Localization of shed PLA₂R within the granular deposits in the subepithelial area thus indicates high likelihood of primary MN, rather than a secondary etiology.

Light microscopy: The earliest stages may have no alterations visible by light microscopy. The early light microscopy lesion consists only of prominent glomerular basement membranes (stage I). In early MN, small deposits may be visualized as a pinpoint “hole” on Jones silver stain, which does not stain the deposits. Ongoing glomerular basement membrane matrix reaction to the deposits results in a “spike” appearance resulting from projections of basement membrane matrix out towards the urinary space on silver stain (stage II). Segmental sclerosis and tubulointerstitial fibrosis may ensue with progressive disease.

Immunofluorescence microscopy: Granular capillary wall deposits of polyclonal immunoglobulin G (IgG; often IgG4 dominant in primary MN), with variable C3, and corresponding positive granular capillary wall PLA₂R staining (performed on formalin-fixed pronase-digested biopsy sections).

Electron microscopy: Extensive foot process effacement and subepithelial deposits with increasing matrix spike reaction with advancing disease. This matrix reaction can ultimately envelop the deposits, resulting in a laddering appearance (stage III).

Deposits may become rarified in chronic disease (stage IV). Mesangial deposits are absent or rare in primary MN.

Etiology/Pathogenesis

Primary MN is caused by autoantibodies to PLA₂R in about 70% of patients. The inciting event(s) causing this autoantibody remain largely unknown. Some patients with MN initiated by nonsteroidal anti-inflammatory drugs may also show PLA₂R positivity. Patients with primary MN usually have circulating anti-PLA₂R antibodies that bind to the PLA₂R on podocytes. The immune complexes formed in situ are shed into the subepithelial area, fix complement, and injure podocytes, resulting in injury, effacement, and marked proteinuria.

Differential Diagnosis

A membranous pattern of injury with spike formation may be seen in other diseases with capillary wall deposits. These typically are PLA₂R negative, and show mesangial deposits and other distinctive features: proliferative glomerulonephritis with monoclinal deposits (diagnosed by the clonal staining with only κ or only λ light chains); amyloid or fibrillary glomerulonephritis (diagnosed by characteristic fibrils shown by electron microscopy); membranous lupus nephritis (“full-house” staining, ie, with IgG, IgA, IgM, C3, and C1q, mesangial deposits, and reticular aggregates); or other secondary causes of MN.

Key Diagnostic Features

- Subepithelial/intramembranous deposits, with spikes/holes appearance of glomerular basement membrane on silver stain
- Granular capillary wall polyclonal IgG and C3 with extensive foot process effacement

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Support: None.

Financial Disclosure: The authors declare that they have no relevant financial interests.

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Am J Kidney Dis. 66(3):e15-e17.

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0272-6386

<http://dx.doi.org/10.1053/j.ajkd.2015.07.006>

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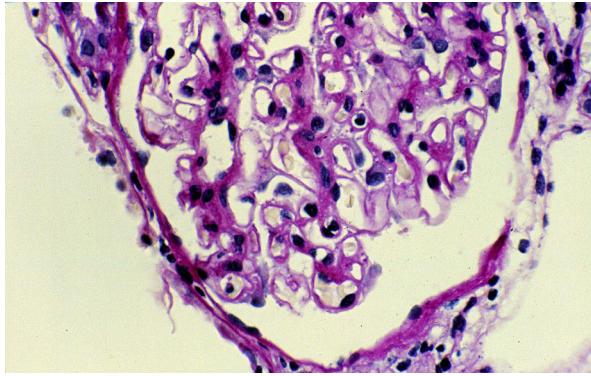


Figure 1. Stage I membranous nephropathy shows slightly prominent capillary walls that appear more rigid than normal; however, deposits cannot be directly visualized (periodic acid–Schiff stain). Reproduced with permission from *AJKD* 31(3):e1.

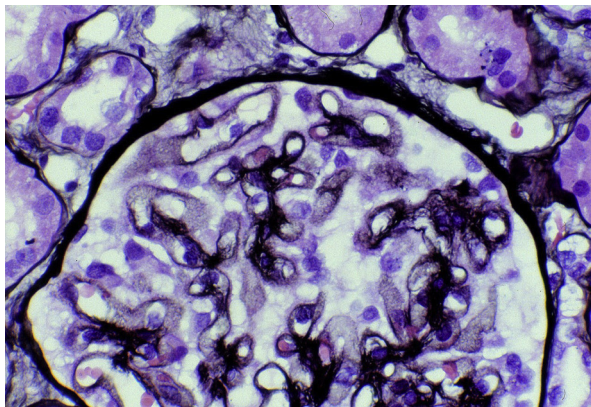


Figure 2. Membranous nephropathy with thickened capillary walls with the appearance of numerous pinpoint “holes” in tangential sections, indicating deposits that did not stain with the Jones silver stain. Well-developed spikes around deposits are not present in this early stage II membranous nephropathy. Reproduced with permission from *AJKD* 31(3):e1.

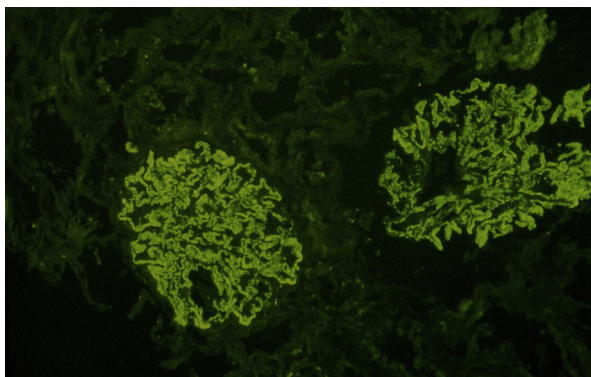


Figure 3. Diffuse granular capillary wall immunoglobulin G (IgG) deposits in membranous nephropathy (immunofluorescence microscopy, IgG). Reproduced with permission from *AJKD* 31(3):e1.

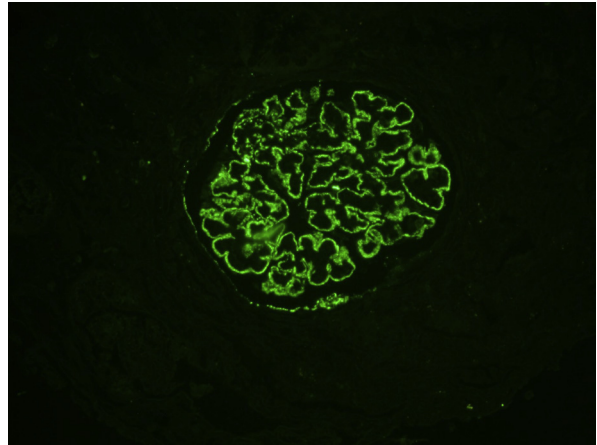


Figure 4. Diffuse granular capillary wall staining for phospholipase A₂ receptor (PLA₂R). PLA₂R staining is positive in approximately 70% of patients with clinically idiopathic membranous nephropathy (immunofluorescence microscopy, PLA₂R).

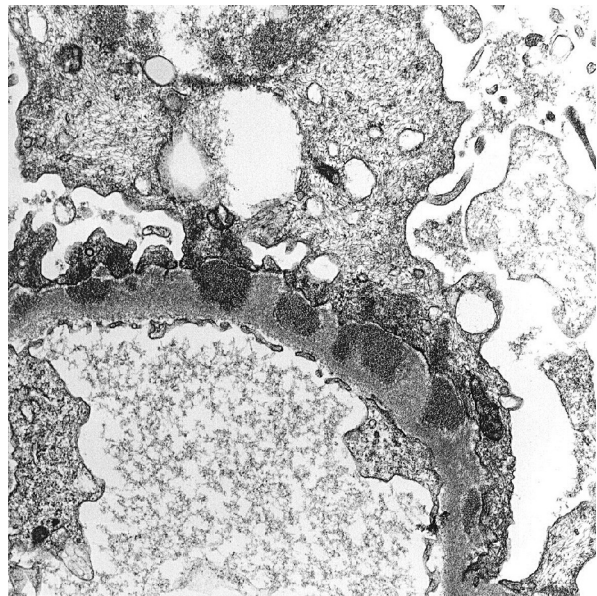


Figure 5. Stage II membranous nephropathy with medium-sized subepithelial deposits and basement membrane reaction, which give rise to the appearance of spikes on Jones silver stain. Podocytes show effaced foot processes (electron microscopy). Reproduced with permission from *AJKD* 31(3):e1.

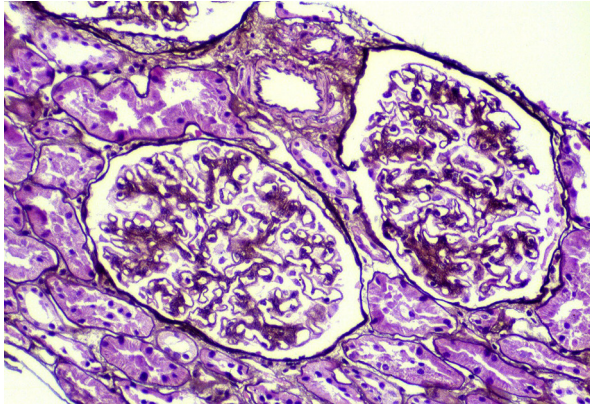


Figure 6. Morphological clues indicate that this membranous nephropathy is secondary rather than idiopathic, including mesangial expansion and mesangial deposits (Jones silver stain). Reproduced with permission from *AJKD* 31(3):e1.



Figure 7. Secondary membranous nephropathy with numerous subepithelial deposits and mesangial deposits (electron microscopy). Reproduced with permission from *AJKD* 31(3):e1.